Data Analysis with Python

Week 6: Pitches and Practice Project

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Plan for today

Student pitches and sample project

- Intro (10 min)
- Pitches ~3 min per project (45 min)
- Break (15 min)
- Sample project (45 min)
  - Git repo setup and folder structure (15 min)
  - Basic Streamlit app + deployment (15 min)
  - Organizing content within app (15 min)

Progress of 6 weeks

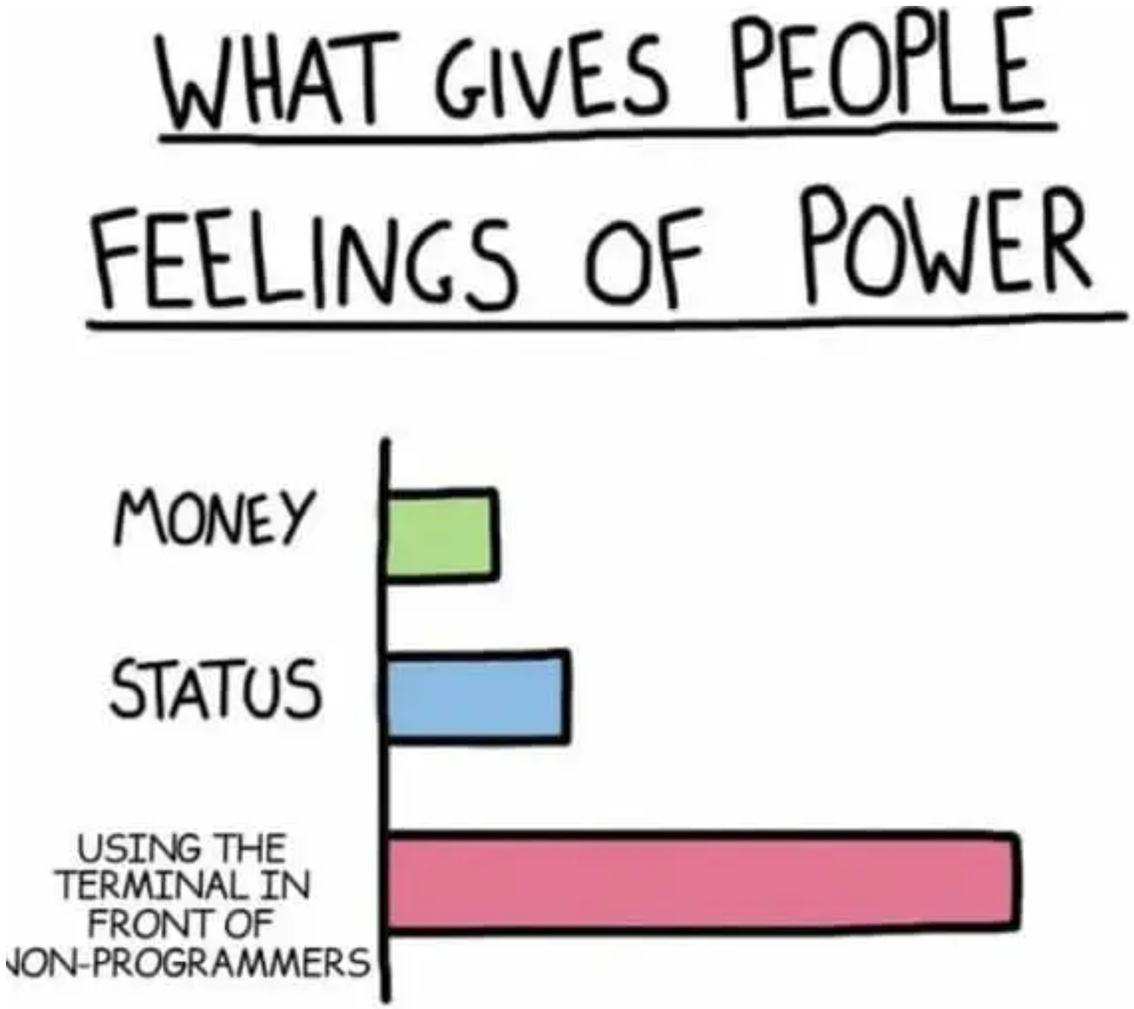
Data Analysis with Python (from zero to hero)

- Weeks 1-2: programming fundamentals with Python
  - Object types, conditional statements, loops, and functions
- Weeks 3-4: data wrangling with Pandas
  - Data cleaning and reshaping
- Weeks 5-6: data analysis and project work
  - Quiz, Aleks (NVIDIA) presentation, visualization, sample project
- Last meeting: project presentations (KOL-H-309)

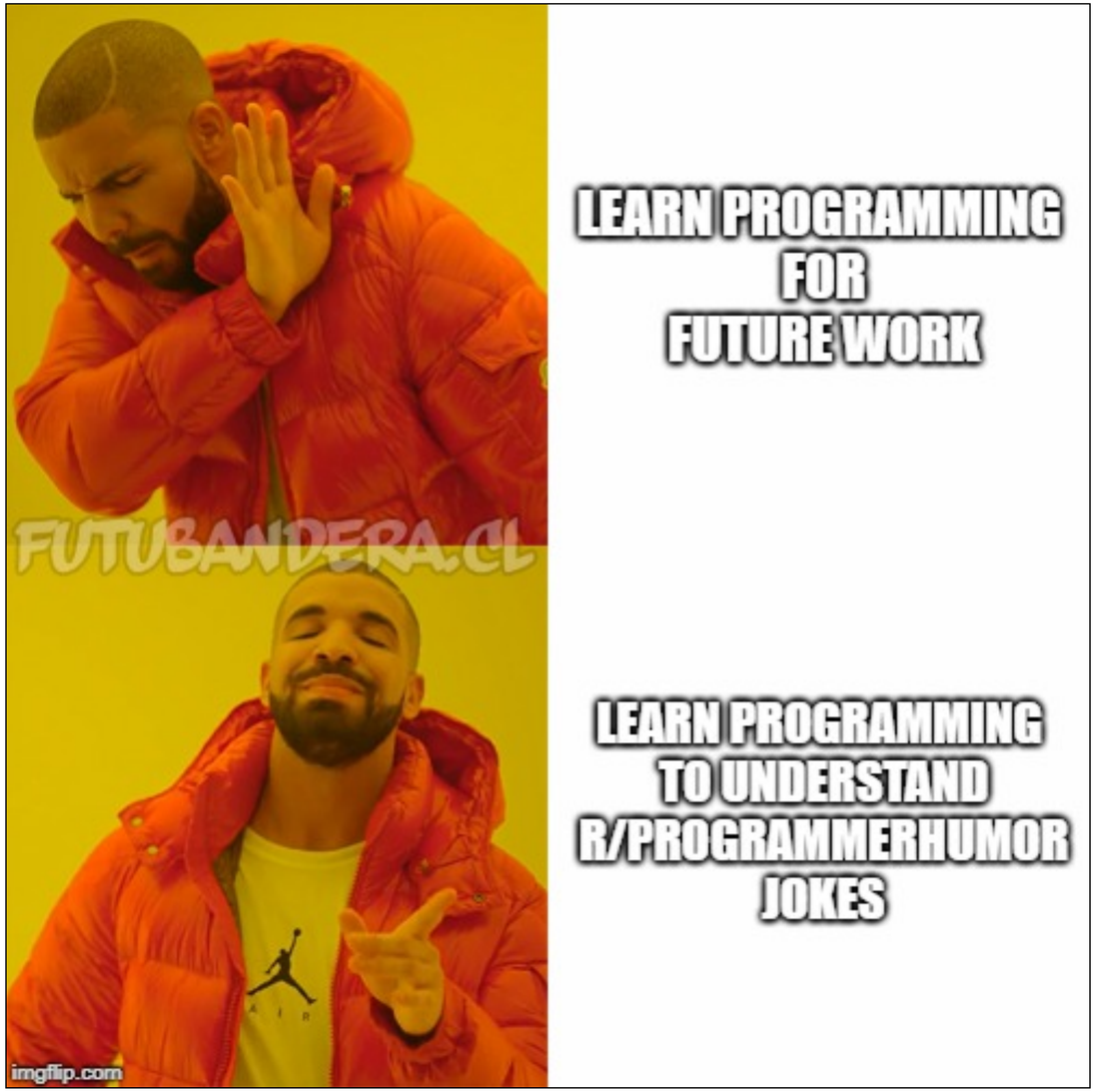
Intro to programming

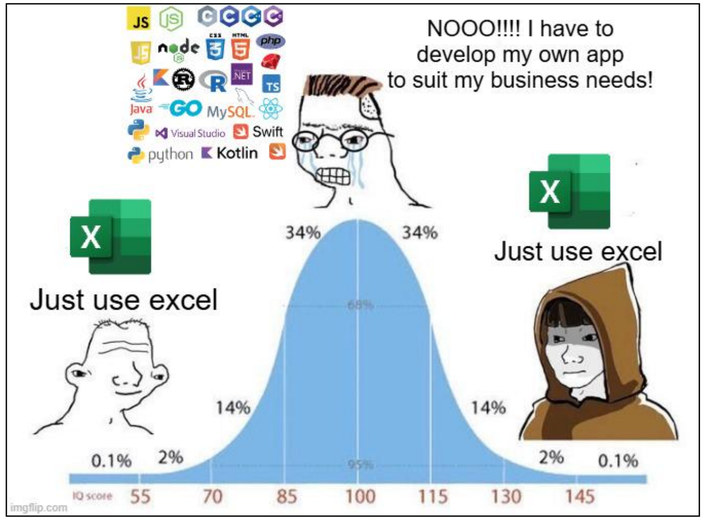
Where to start?

- Terminal
- VS Code
- Git and GitHub
- HTML and GitHub pages
- Python and Streamlit
- Jupyter Notebook

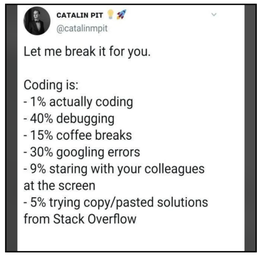


r/programmingmemes; r/ProgrammerHumor



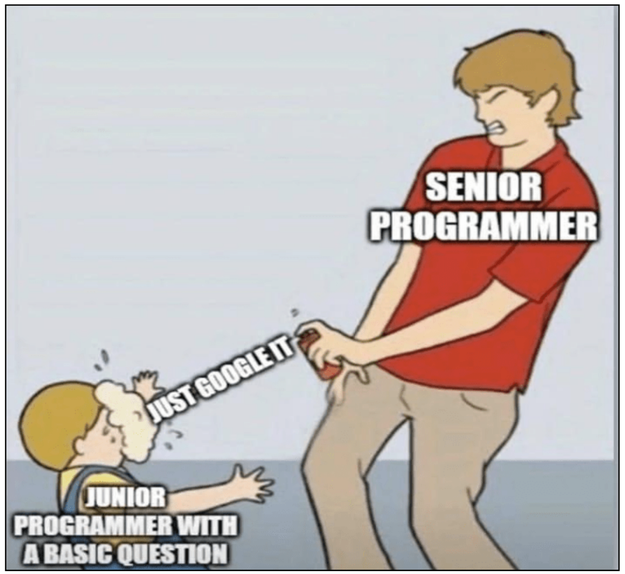












Course Feedback

Accelerated style, agency, and personalisation

What worked well?

- Individual learning and weekly self-assessment
- High-level overview presentations and choice of topics
- Weekly in-class activities
- Fast pace and project-oriented from early on
- Progress evaluation structure (weekly + quiz + pitches + final project)
- Friday emails

Course Feedback

Accelerated style, agency, and personalisation

What could be improved?

- Flipped classroom - suggested videos before the class, not after
- More specific homework exercises
- Installation guide before the first class
- Start with individual GitHub repos (no Playground) for weeks 1-4

Conflicted about: too much Git; simplified setup (Anaconda instead of VS Code); more in-class coding / assistance

Final Presentations (Dec 4, 12-16:00, KOL-H-309)

Project rubric and goals



- Baseline grades ~4.5
- Project submitted and presented, runs without errors: +0.25
- Main steps (wrangling, analysis, visualization): +0.25
- Git history (regular commits): +0.25
  - Suggestion: self-assigned checkpoints, e.g., Oct 29, Nov 12, Nov 26
- Bonus points for deep thinking and technical complexity: +0.25-0.75

Please reach out for help to Gustav or me (especially early on)

Final Report (Dec 10, 23:59, OLAT)

Aim for 2-3 pages, no AI please

- Admin bookkeeping (names, link to GitHub, presentation attached)
- Topic, main questions (up to 3-4)
- Main takeaways (up to 1-2 per question)
- Technical details
  - Data used (raw data links) and data cleaning steps (high level)
  - Data analysis steps, bonus features (e.g., what makes you standout)
- Personal contribution split
- Human and AI effort disclaimer

Homework

Homework (~5 hours)

Setup for the final project

- Create a final project GitHub repo (~1 hour)
  - Use personal GitHub, invite partner(s), Gustav, and me as collaborators
- Create a basic app.py and deploy on Streamlit Cloud (~1 hr)
- Design a layout with st.columns and st.containers (~1 hr)
  - Think about multipage setup
- Upload data and display some analysis (~2 hr)
  - Basic wrangling + metadata or summary stats + 2-3 graphs

Project pitches

Sample project (record Zoom)

Sample project
Key checkpoints

- Folder structure - app, data, drafts, scripts, README.md, LICENSE
- Basic Streamlit app.py and deployment - https://share.streamlit.io/
  - Localhost vs deployed version, relative paths
  - Requirements.txt (environment, containers)
- Organizing content within the app
  - Headers, columns, and containers
  - Multipage setup
- Environment/secret variables (!!)
- Database connection (bonus)

Sample project

Agentic engineering

- Brainstorm ideas, define the problem, envision the solution
- Roadmap (PRD - Product Requirements Doc), priorities, steps
- Define agent skills / tools / hooks based on the roadmap
- Split roadmap across multiple agents by domain
- Launch agents in parallel, guide them with skills
- Must write logs, docs, and agent handoffs
- Verify output, update plans, iterate